Assignment 2 — Federated Learning for Healthcare Analytics

Course: ITAI 4376 — AI Applications
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Problem Statement

Healthcare institutions hold valuable patient data that could be used to train powerful predictive models, but sharing that data across institutions raises serious privacy and regulatory concerns. This assignment addressed the question: how can machine learning models be trained on distributed healthcare data without centralizing sensitive patient information?

Approach and Methods

The Neural Spark team designed and implemented a privacy-preserving federated learning prototype for healthcare analytics. Rather than aggregating raw patient data in a central server, our system trained local models at each simulated client node and shared only model weights with a central aggregation server — ensuring that no individual patient record left its source environment.

The system was built using TensorFlow Federated for the distributed learning framework, Flask for the backend API, and a React frontend for monitoring and visualization. Synthetic electronic health record data generated by the Synthea platform served as the training dataset, allowing realistic healthcare scenarios to be simulated without using real patient information.

The federated averaging algorithm was used to aggregate model updates from participating clients, and the system was evaluated on metrics including model accuracy, communication efficiency, and privacy preservation.

Results and Interpretation

The prototype demonstrated that federated learning can achieve competitive model performance compared to centralized training while maintaining strict data locality. The React dashboard provided real-time visibility into training rounds, client participation, and model convergence, making the system accessible to non-technical stakeholders.

The use of Synthea data allowed the team to validate the system against realistic clinical scenarios including patient readmission prediction and chronic disease risk stratification. Results indicated that the federated approach introduced only a modest accuracy trade-off relative to centralized training, while providing substantially stronger privacy guarantees.

Key Takeaway

Federated learning represents a practical and principled solution to one of the most persistent barriers in healthcare AI — the tension between data utility and patient privacy. This project demonstrated that it is possible to build powerful predictive systems that respect both regulatory requirements and patient rights.

Technologies Used

- TensorFlow Federated
- Flask (Python backend)
- React (frontend dashboard)
- Synthea (synthetic EHR data generation)
- GitHub for version control and portfolio documentation